Forecast Error Map EDA

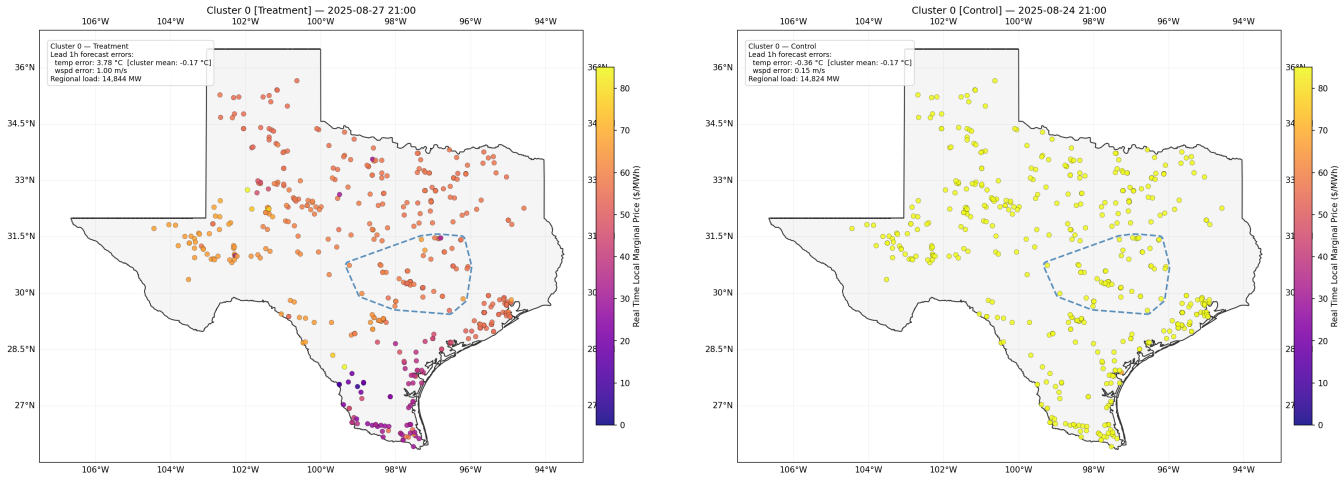
Temperature errors, lead 1h (HRRR)

March 10, 2026

Configuration

Weather variable	`temp` (temperature, °C)
Lead time	1h
Model	HRRR
Months	2025-01 to 2025-08
Clusters	9

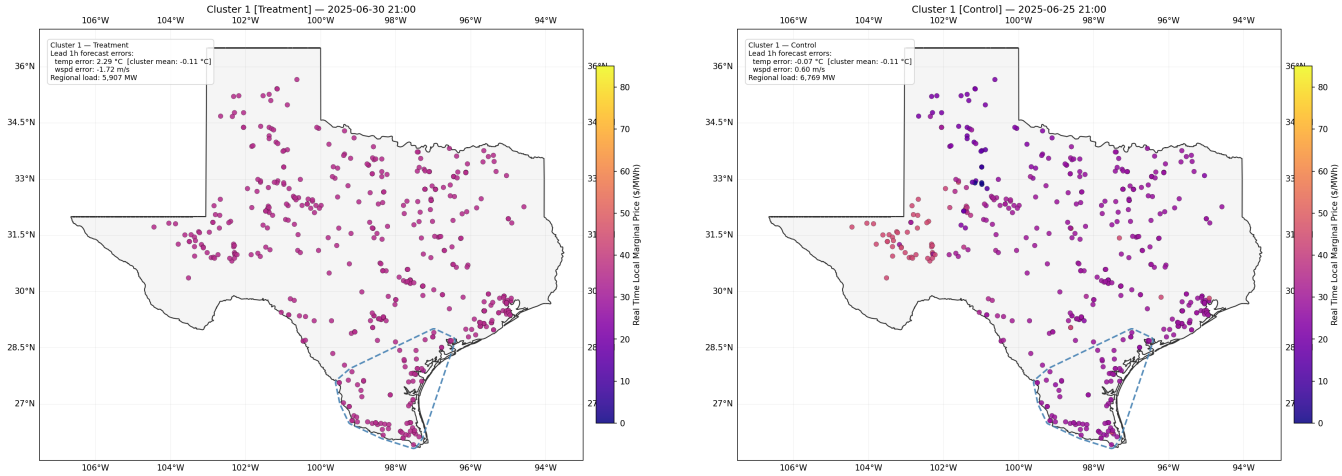
For each cluster, the **treatment** map shows the hour with the largest absolute mean temperature forecast error in that cluster. The **control** map shows a nearby hour (within 14 days) at the same hour of day with temperature error close to the cluster mean. Node colors show real-time LMP ($/MWh); the color scale is shared between treatment and control within each cluster. Annotations report the temperature error for that hour alongside both wind speed and temperature errors for reference.



(a) Treatment (large error)

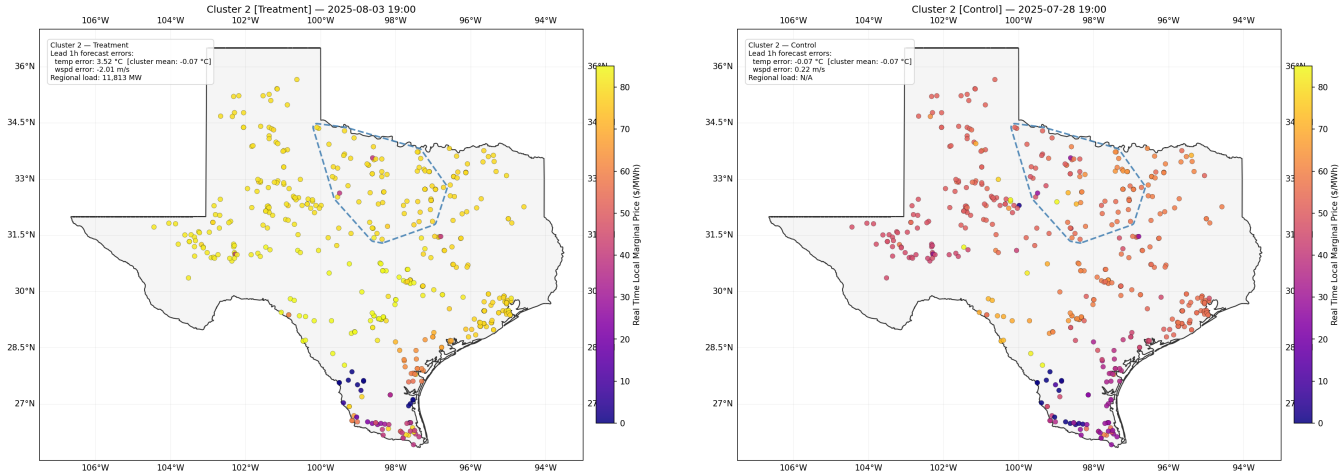
(b) Control (near-mean error)

Figure 1: Cluster 0: LMP maps for treatment vs. control hours (temperature error, lead 1h).



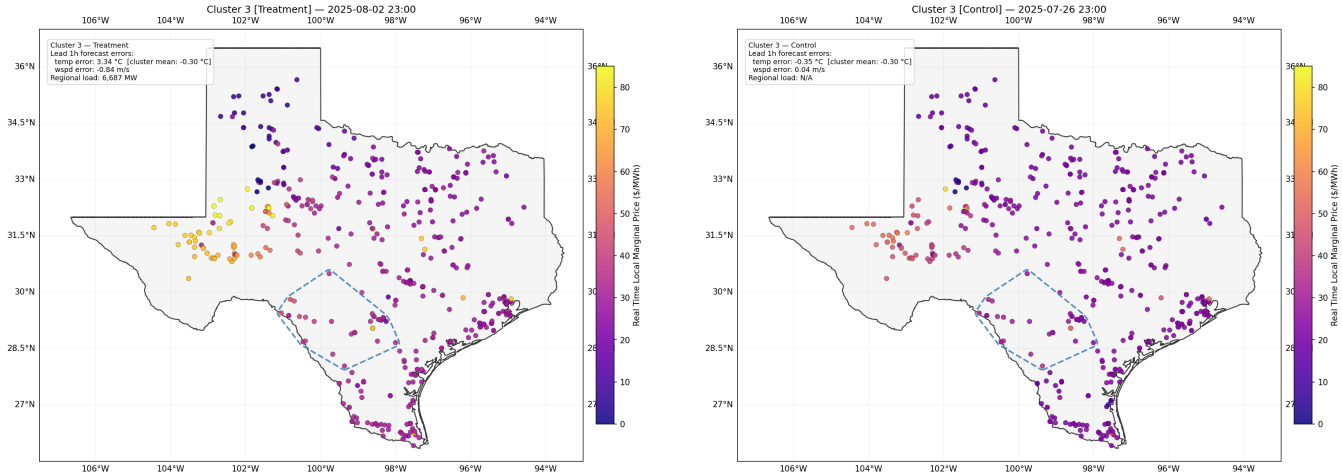
(a) Treatment (large error) (b) Control (near-mean error)

Figure 2: Cluster 1: LMP maps for treatment vs. control hours (temperature error, lead 1h).



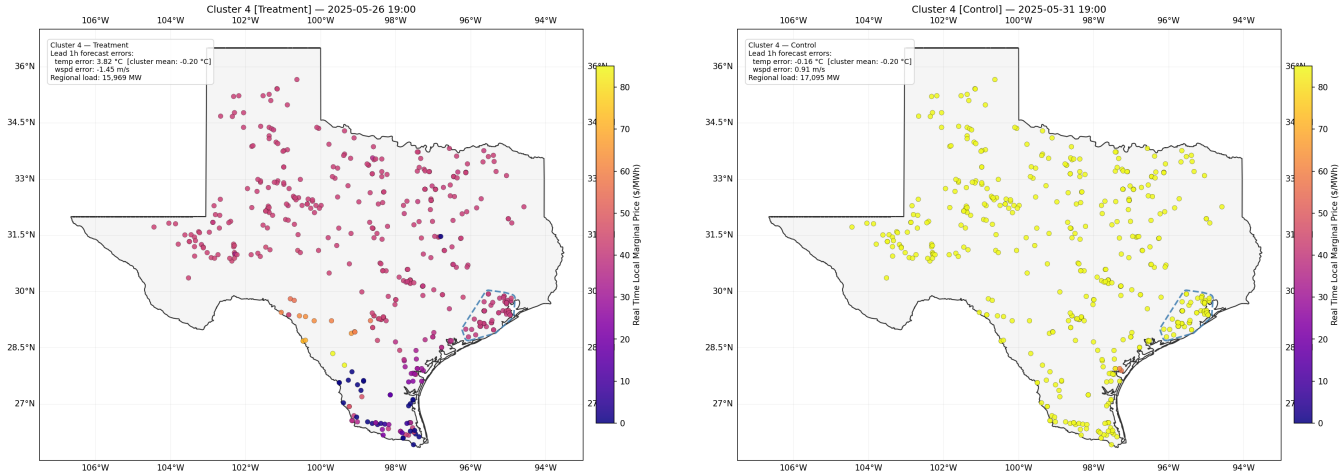
(a) Treatment (large error) (b) Control (near-mean error)

Figure 3: Cluster 2: LMP maps for treatment vs. control hours (temperature error, lead 1h).



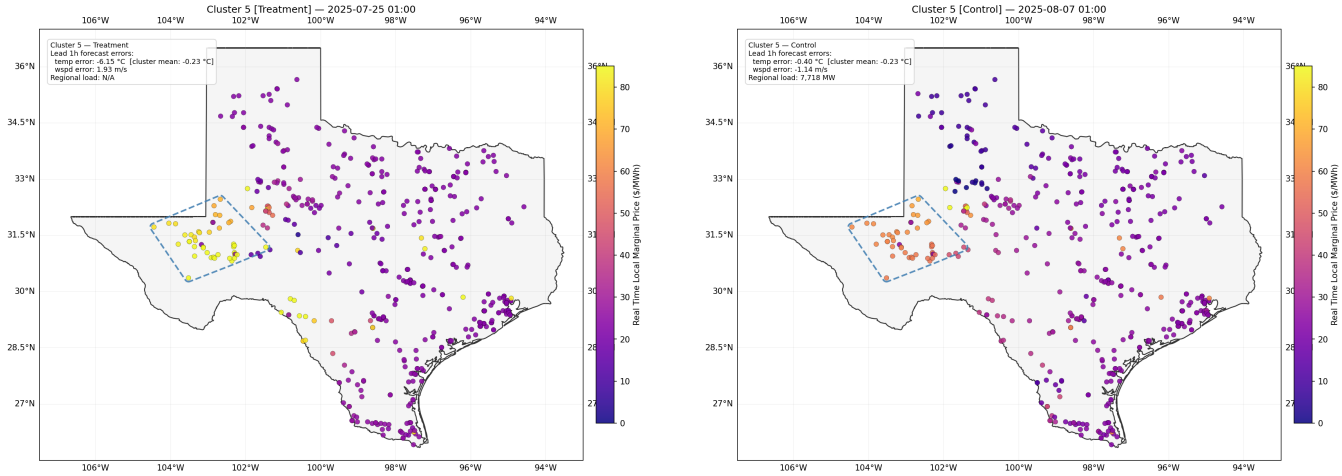
(a) Treatment (large error) (b) Control (near-mean error)

Figure 4: Cluster 3: LMP maps for treatment vs. control hours (temperature error, lead 1h).



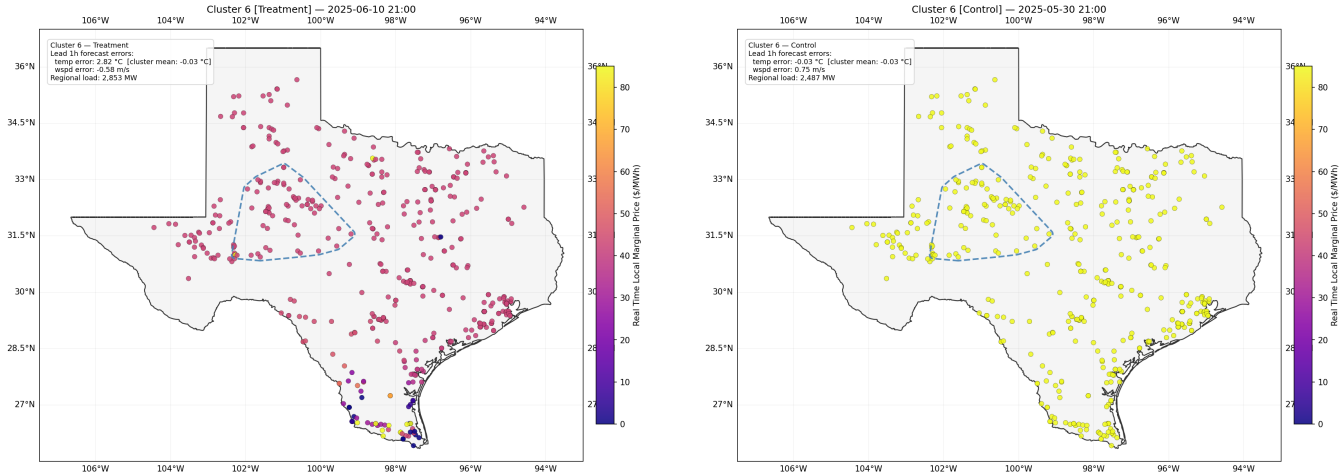
(a) Treatment (large error) (b) Control (near-mean error)

Figure 5: Cluster 4: LMP maps for treatment vs. control hours (temperature error, lead 1h).



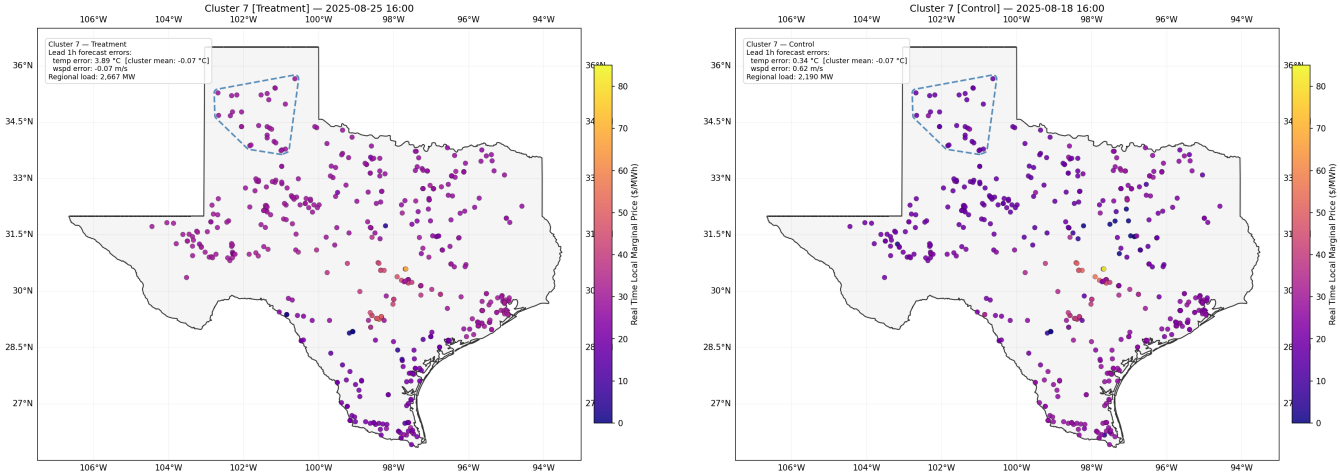
(a) Treatment (large error) (b) Control (near-mean error)

Figure 6: Cluster 5: LMP maps for treatment vs. control hours (temperature error, lead 1h).



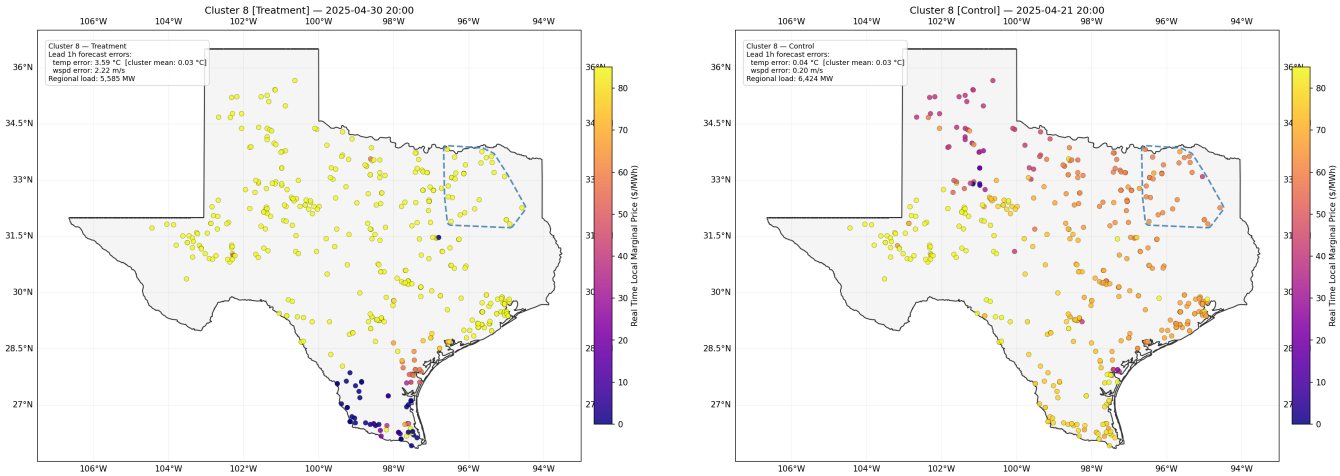
(a) Treatment (large error) (b) Control (near-mean error)

Figure 7: Cluster 6: LMP maps for treatment vs. control hours (temperature error, lead 1h).



(a) Treatment (large error) (b) Control (near-mean error)

Figure 8: Cluster 7: LMP maps for treatment vs. control hours (temperature error, lead 1h).



(a) Treatment (large error) (b) Control (near-mean error)

Figure 9: Cluster 8: LMP maps for treatment vs. control hours (temperature error, lead 1h).